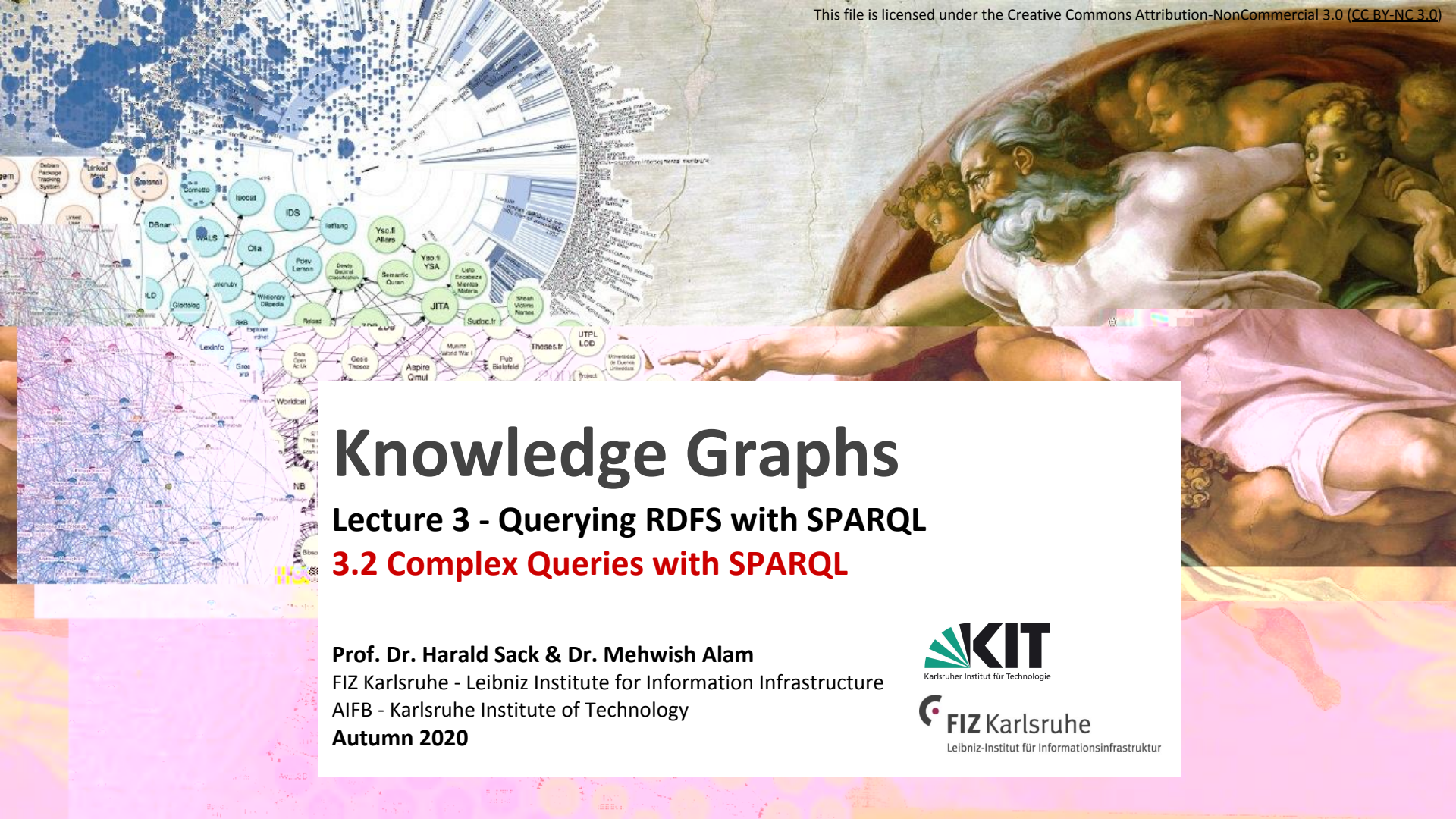




Knowledge Graphs

Lecture 3 - Querying RDFS with SPARQL

3.2 Complex Queries with SPARQL

Prof. Dr. Harald Sack & Dr. Mehwish Alam

FIZ Karlsruhe - Leibniz Institute for Information Infrastructure

AIFB - Karlsruhe Institute of Technology

Autumn 2020



Knowledge Graphs

Lecture 3: Querying RDF(S) with SPARQL

3.1 How to Query RDF(S)

Excursion 2: DBpedia Knowledge Graph

Excursion 3: Wikidata Knowledge Graph

3.2 Complex Queries with SPARQL

3.3 More Complex SPARQL Queries

3.4 SPARQL Subqueries and Property Paths

3.5 RDF Databases

3.6 SPARQL is more than a Query Language

The Semantic Web Technology Stack (not a piece of cake...)

Most apps use only a subset of the stack

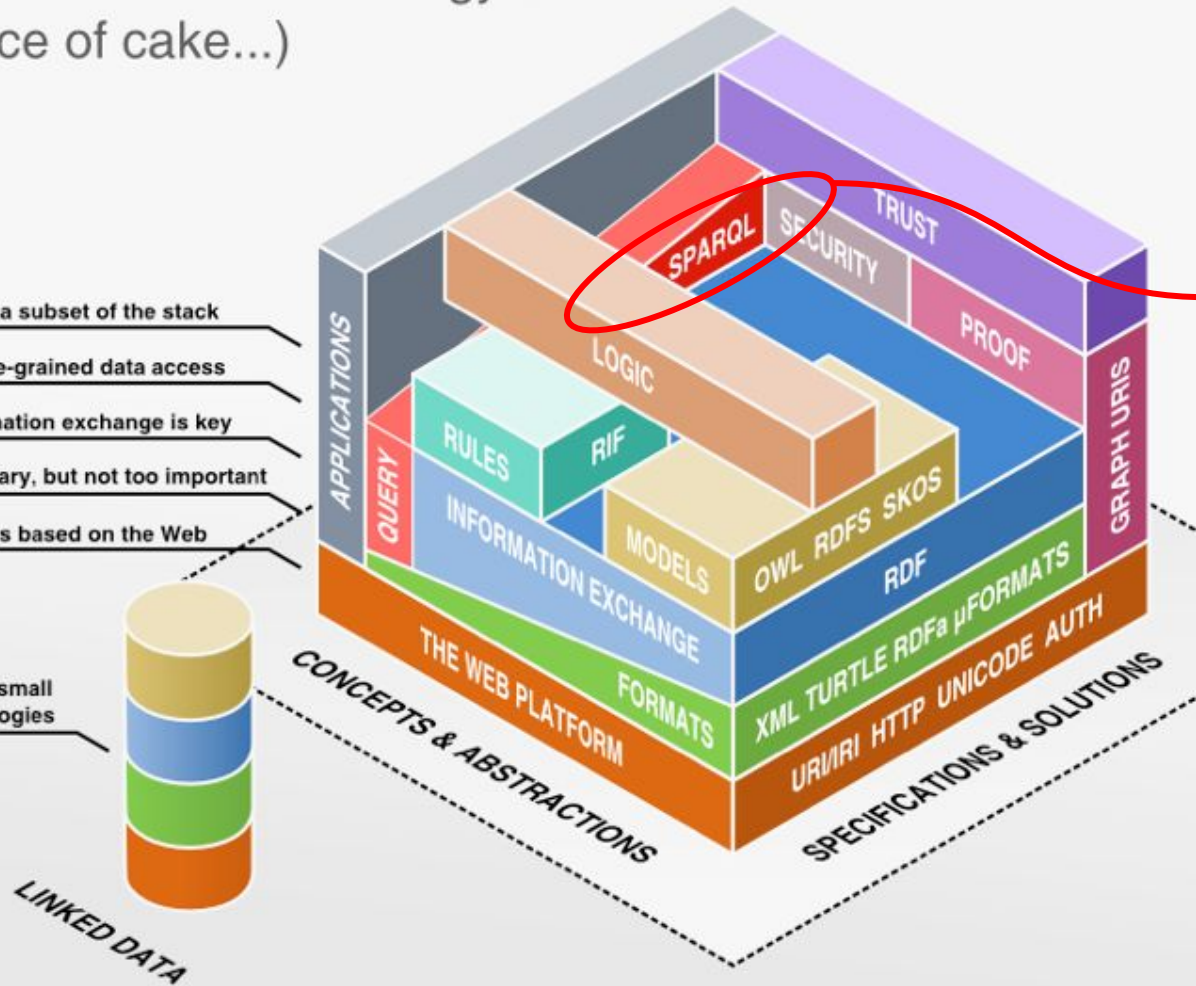
Querying allows fine-grained data access

Standardized information exchange is key

Formats are necessary, but not too important

The Semantic Web is based on the Web

Linked Data uses a small
selection of technologies



USE
SPARQL
and
QUERY
ON

SPARQL Filter Constraints

```

: <http://dbpedia.org/resource/>
rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
rdfs: <http://www.w3.org/2000/01/rdf-schema#>
dbo: <http://dbpedia.org/ontology/>
dct: <http://purl.org/dc/terms/>
dbc: <http://dbpedia.org/resource/Category:>

?author_name ?title
<http://dbpedia.org/>
{
  ?author rdf:type dbo:Writer .
  ?author rdfs:label ?author_name
    ( (?author_name)="en") .
  ?work dbo:author ?author .
    ?work rdfs:label ?title .
    ( (?title)="en")
  ?work dct:subject dbc:Environmental_fiction_books .
}
100

```

- Example:
Search for **authors**
and their **books**, filter
results for **English**
labels and
Environmental Fiction
Books and limit the
results to the **first**
100



[query SPARQL endpoint](#)

WIKIDATA Label Language Filtering

```
wd: <http://www.wikidata.org/entity/>
wdt: <http://www.wikidata.org/prop/direct/>
wikibase: <http://wikiba.se/ontology#>
bd: <http://www.bigdata.com/rdf#>
```

```
?authorLabel ?bookLabel ?date
```

```
?book wdt:P31 wd:Q47461344 .
```

```
?book wdt:P50 ?author .
```

```
?book wdt:P921 wd:Q7942 .
```

```
?book wdt:P577 ?date .
```

```
wikibase:label
```

```
bd:serviceParam wikibase:language "en"
```

Instance of written work

author

main subject global warming

publication date

- Example:
Search for authors
and their books,
filter results for
English labels and
Books on Global
warming



[query SPARQL endpoint](#)

WIKIDATA Label Language Filtering

```
wd: <http://www.wikidata.org/entity/>
wdt: <http://www.wikidata.org/prop/direct/>
wikibase: <http://wikiba.se/ontology#>
bd: <http://www.bigdata.com/rdf#>
```

```
?authorLabel ?bookLabel ?date
```

```
?book wdt:P31 wd:Q47461344 .
?book wdt:P50 ?author .
?book wdt:P921 wd:Q7942 .
?book wdt:P577 ?date .
```

```
wikibase:label
bd:serviceParam wikibase:language "en"
```

wikidata specific
label service

- Example:
Search for authors
and their books,
filter results for
English labels and
Books on Global
warming



[query SPARQL endpoint](#)

3. Querying RDF(S) with SPARQL / 3.2 Complex Queries with SPARQL

- Example:
Search for authors and their books, filter results for English labels and
Books on global warming

 Wikidata Query Service
 Examples
Help
More tools
English

```

1 SELECT ?authorLabel ?bookLabel ?date
2 WHERE {
3   ?book wdt:P31 wd:Q47461344 . # instance of (P31) written work (Q47461344)
4   ?book wdt:P50 ?author .      # author (P50)
5   ?book wdt:P921 wd:Q7942 .    # main subject (P921) global warming (Q7942)
6   ?book wdt:P577 ?date .       # publication date (P577)
7   SERVICE wikibase:label
8   { bd:serviceParam wikibase:language "en" }
9 }
10

```

8 results in 580 ms
Code
Download
Link

authorLabel	bookLabel	date
Al Gore	An Inconvenient Truth	1 January 2006
Alain Grandjean	It's Now! 3 Years to Save the World	1 January 2009
Jean-Marc Jancovici	It's Now! 3 Years to Save the World	1 January 2009
Björn Lomborg	Cool It: The Skeptical Environmentalist's Guide to Global Warming	1 January 2007
Chris Goodall	Ten Technologies to Fix Energy and Climate	13 November 2008

[query SPARQL endpoint](#)

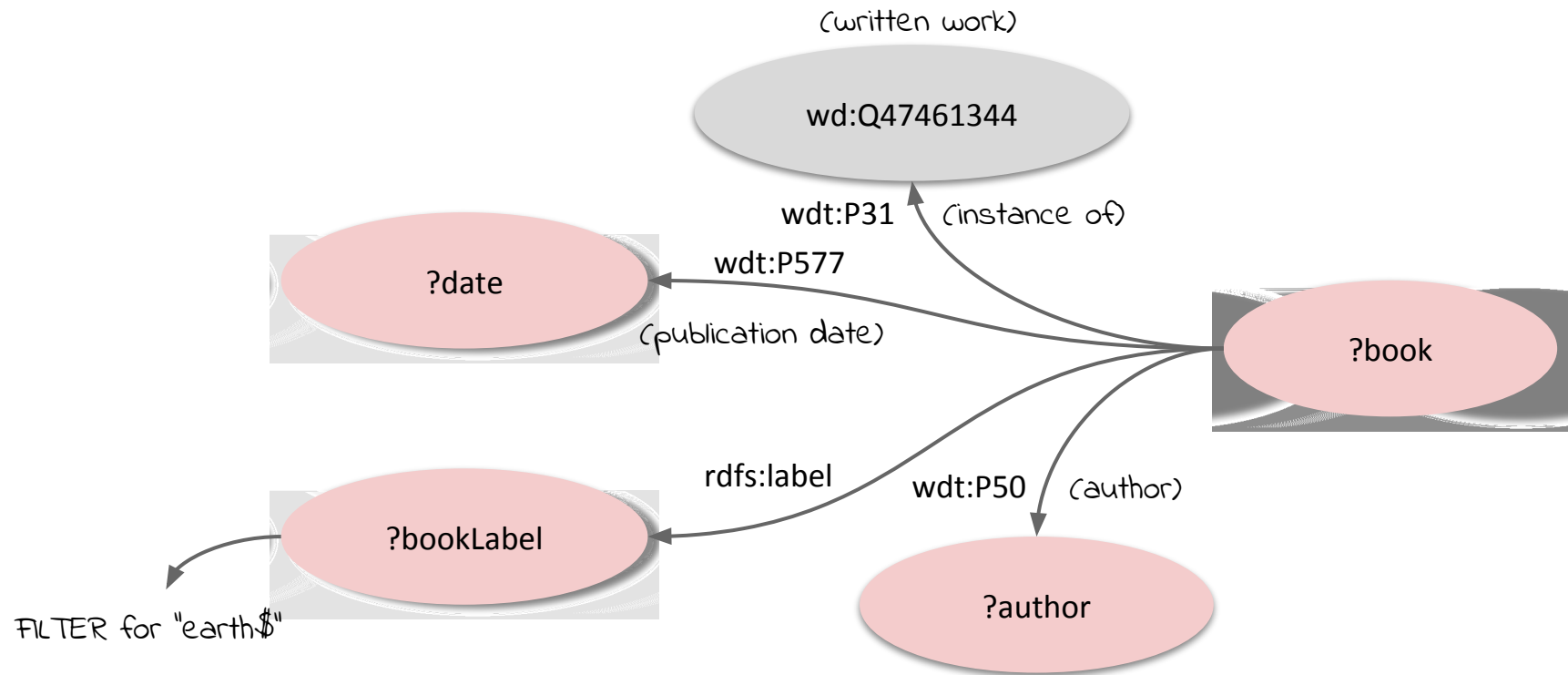
More SPARQL Operators

- Logical connectives **(AND)** and **(OR)** for xsd:boolean
- Comparison operators `<`, `<=`, `>`, `>=`, and `=` for numeric datatypes, xsd:dateTime, xsd:string, and xsd:boolean
- Comparison operators `is` and `is not` for other datatypes
- Arithmetic operators `+`, `-`, `*`, and `div` for numeric datatypes
- and in addition:
 - `not`
 - `distinct`
 - `order by`

or

SPARQL Filter Constraints

- what book titles end with the word "earth" sorted by publication date?



SPARQL Filter Constraints

- what book titles end with the word "earth" sorted by publication date?



```
wd: <http://www.wikidata.org/entity/>
wdt: <http://www.wikidata.org/prop/direct/>
rdfs: <http://www.w3.org/2000/01/rdf-schema#>
```

```
?authorLabel ?bookLabel ?date
```

```
?book wdt:P31 wd:Q47461344 .
```

```
?book wdt:P50 ?author .
```

```
?book wdt:P577 ?date
```

```
?book rdfs:label ?bookLabel
```

```
?author rdfs:label ?authorLabel
```

```
?date
```

string

regular
expression

flags

- With **FILTER**
REGEX, regular
expressions can
be filtered

[query SPARQL endpoint](#)

3. Querying RDF(S) with SPARQL / 3.2 Complex Queries with SPARQL

- what book titles end with the word "earth" sorted by publication date?

Wikidata Query Service

```

1 PREFIX wd: <http://www.wikidata.org/entity/>
2 PREFIX wdt: <http://www.wikidata.org/prop/direct->
3 PREFIX wikibase: <http://wikiba.se/ontology#>
4 PREFIX bd: <http://www.bigdata.com/rdf#>
5
6 SELECT ?authorLabel ?bookLabel ?date
7 WHERE {
8   ?book wdt:P50 ?author .
9   ?book wdt:P577 ?date .
10  ?book rdfs:label ?bookLabel FILTER (LANG(?bookLabel)="en")
11  FILTER regex (?bookLabel,"earth$","i") .
12  ?author rdfs:label ?authorLabel FILTER (LANG(?authorLabel)="en")
13 }
14
15 ORDER BY ?date
16

```

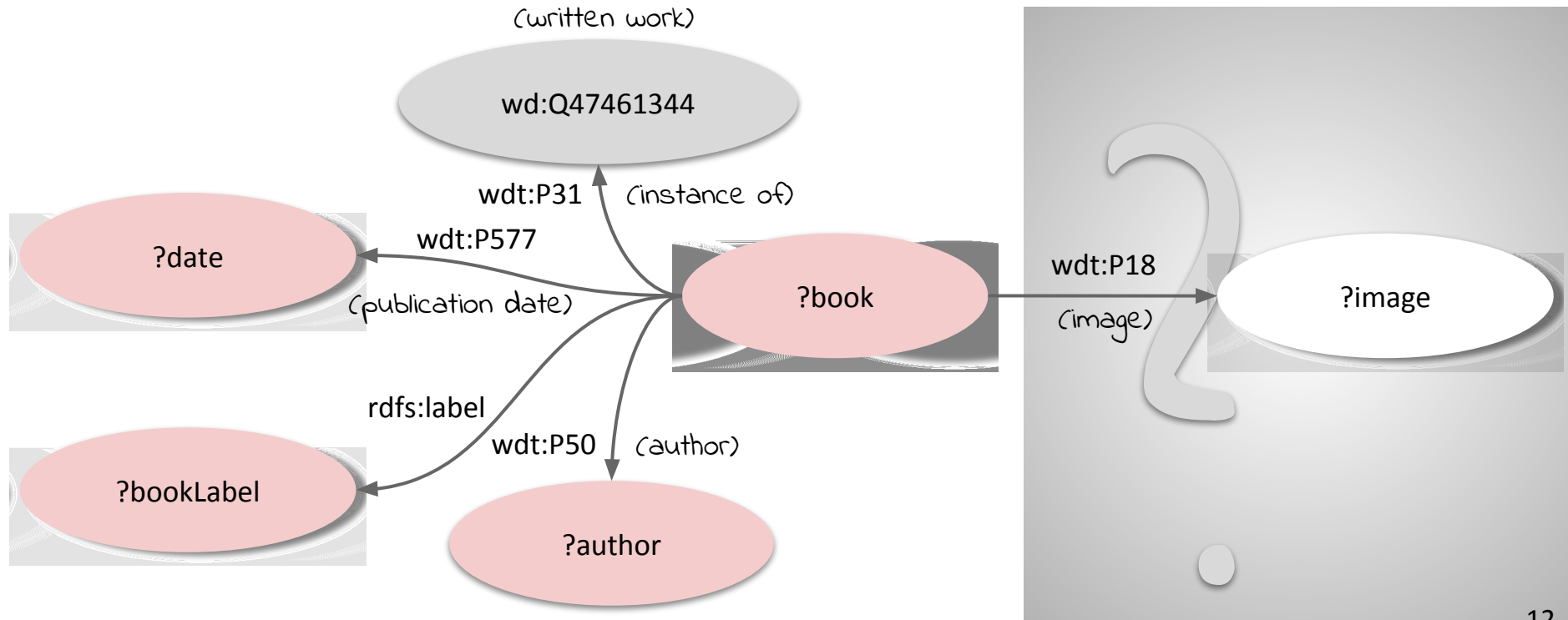
48 results in 9623 ms

authorLabel	bookLabel	date
Charles Dickens	The Cricket on the Hearth	20 December 1845
André Gide	The Fruits of the Earth	1 January 1897
H. G. Wells	The Food of the Gods and How It Came to Earth	1 January 1904
Bruce Marshall	Children of This Earth	1 January 1930
Pearl S. Buck	The Good Earth	2 March 1931
Raymond F. Jones	This Island Earth	1 January 1952
François Bordes	Fleeing Earth	1 January 1960
Arthur Koestler	Scum of the Earth	1 January 1960

[query SPARQL endpoint](#)

SPARQL Filter Constraints

- which book titles end with the word "earth", and, if available, do also have an image?

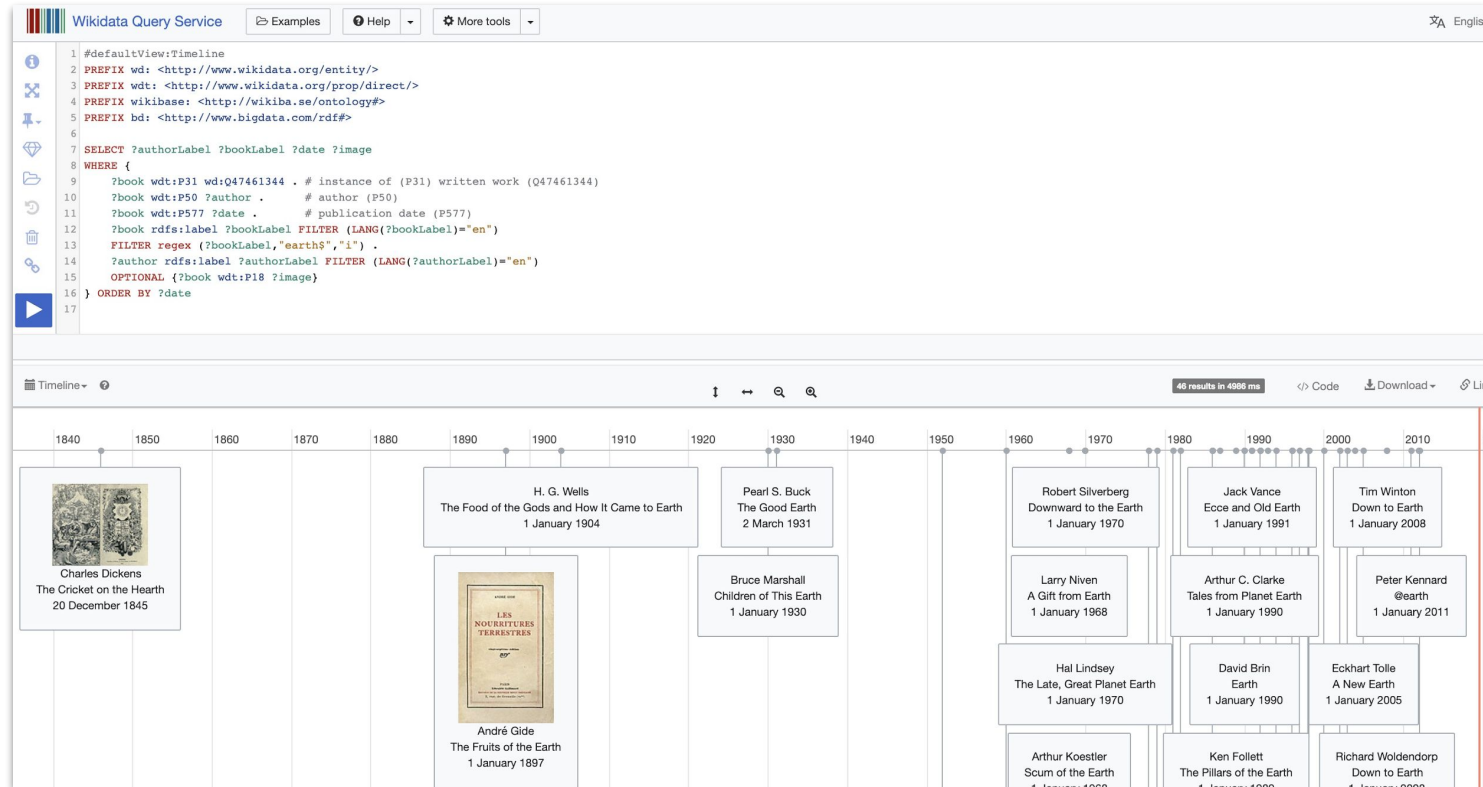


SPARQL Filter Constraints

- Optional selection of graph pattern via **OPTIONAL**

3. Querying RDF(S) with SPARQL / 3.2 Complex Queries with SPARQL

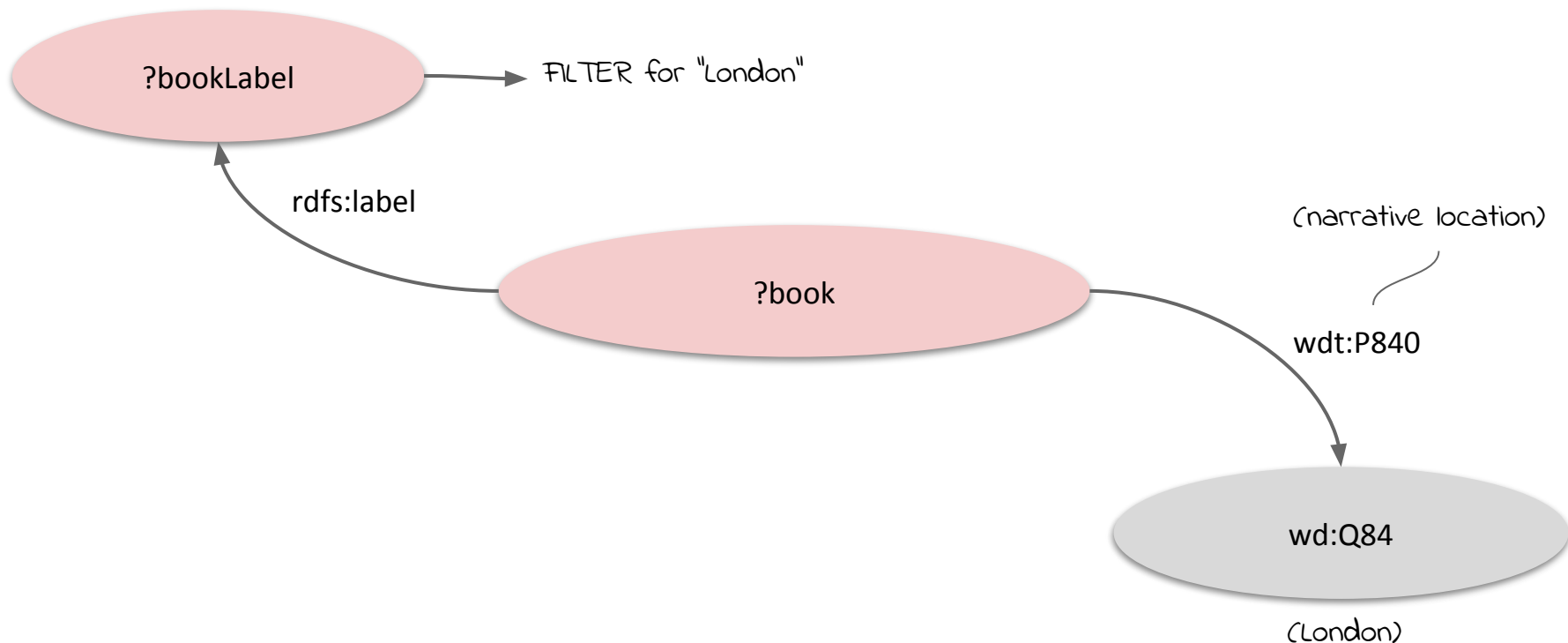
- which book titles end with the word "earth", and, if available, do also have an image?



[query SPARQL endpoint](#)

SPARQL Alternative Results via UNION

- Example: which books mention "London" in their title **or** have London as their narrative location



SPARQL Alternative Results via UNION

- Example: which books mention "London" in their title *or* have London as their narrative location



```

wd: <http://www.wikidata.org/entity/>
wdt: <http://www.wikidata.org/prop/direct/>
wikibase: <http://wikiba.se/ontology#>
bd: <http://www.bigdata.com/rdf#>

?authorLabel ?bookLabel ?book ?date ?image
{
  ?book wdt:P31 wd:Q47461344 .
  ?book wdt:P50 ?author .
  ?book wdt:P577 ?date .
  {
    (?bookLabel, "London", "i") . }
  { ?book wdt:P840 wd:Q84 . }
  wikibase:label { bd:serviceParam wikibase:language "en" }
  ?date
}

```

*logical
disjunction*

- The keyword **UNION** allows for alternatives (logical disjunction)

3. Querying RDF(S) with SPARQL / 3.2 Complex Queries with SPARQL

- Example: which books mention "London" in their title or have London as their narrative location

Wikidata Query Service

```

1 PREFIX wd: <http://www.wikidata.org/entity/>
2 PREFIX wdt: <http://www.wikidata.org/prop/direct/>
3 PREFIX wikibase: <http://wikiba.se/ontology#>
4 PREFIX bd: <http://www.bigdata.com/rdf#>
5
6 SELECT ?authorLabel ?bookLabel ?book ?date
7 WHERE {
8   ?book wdt:P31 wd:Q47461344 . # instance of (P31) written work (Q47461344)
9   ?book wdt:P50 ?author . # author (P50)
10  ?book wdt:P577 ?date . # publication date (P577)
11  {
12    FILTER regex (?bookLabel,"London","i") .
13  }
14  UNION
15  {
16    ?book wdt:P840 wd:Q84 # narrative location (P840) London (Q84)
17  }
18  SERVICE wikibase:label { bd:serviceParam wikibase:language "en" }
19 } ORDER BY ?date
20

```

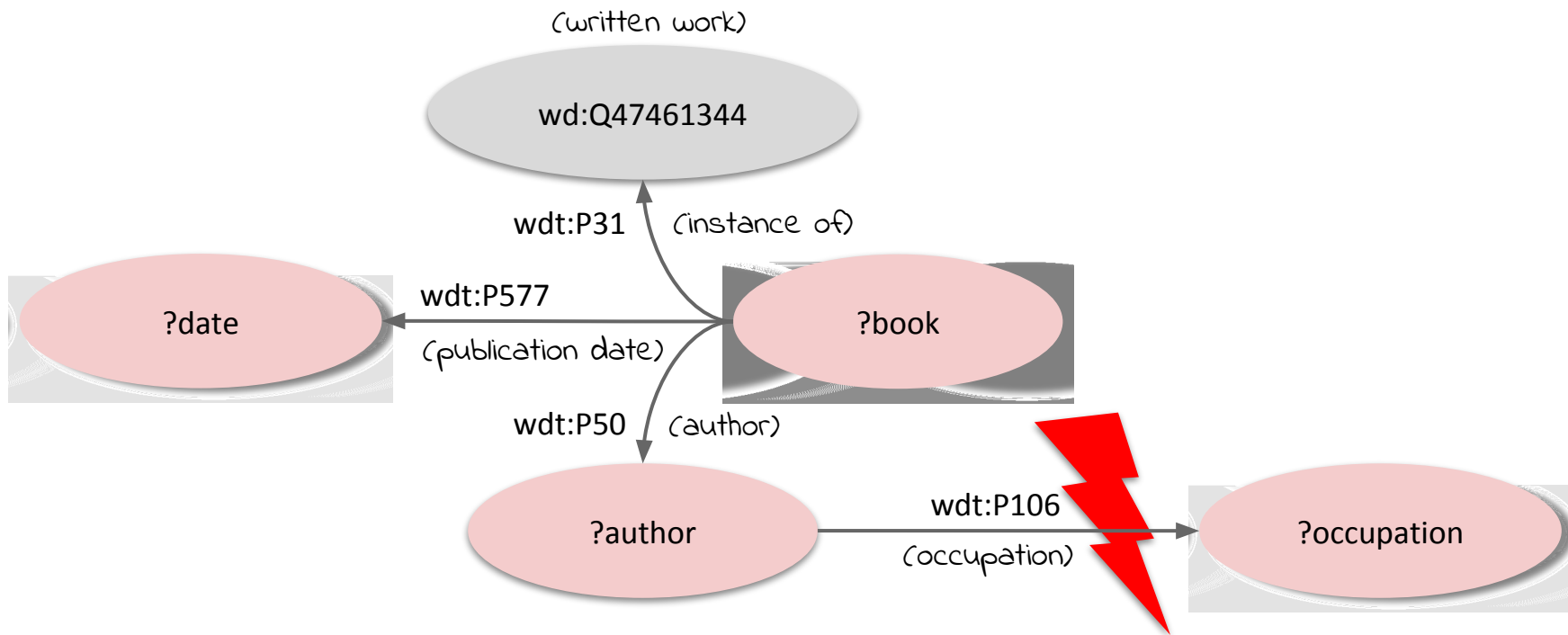
238 results in 1721 ms

authorLabel	bookLabel	book	date
Girolamo Graziani	Il Cromuele	Q wd:Q3792648	1 January 1671
Daniel Defoe	A Journal of the Plague Year	Q wd:Q1215399	1 January 1722
Frances Burney	Cecilia	Q wd:Q3233990	1 July 1782
Charles Dickens	Nicholas Nickleby	Q wd:Q847642	1 January 1839
Charles Dickens	A Christmas Carol	Q wd:Q62879	19 December 1843
Alexandre Dumas	The Three Musketeers	Q wd:Q140527	1 January 1844
Charles Dickens	The Cricket on the Hearth	Q wd:Q825220	20 December 1845
Charles Dickens	A Tale of Two Cities	Q wd:Q308918	1 January 1859
Charles Dickens	Great Expectations	Q wd:Q219552	1 January 1860

[query SPARQL endpoint](#)

SPARQL Negation

- Example: which books are written by authors who **don't** have an occupation ?



SPARQL Negation

- Example: which books are written by authors who **don't have an occupation** ?



*filter query
result for
existence*

SPARQL 1.1 offers several variants for negation:

○

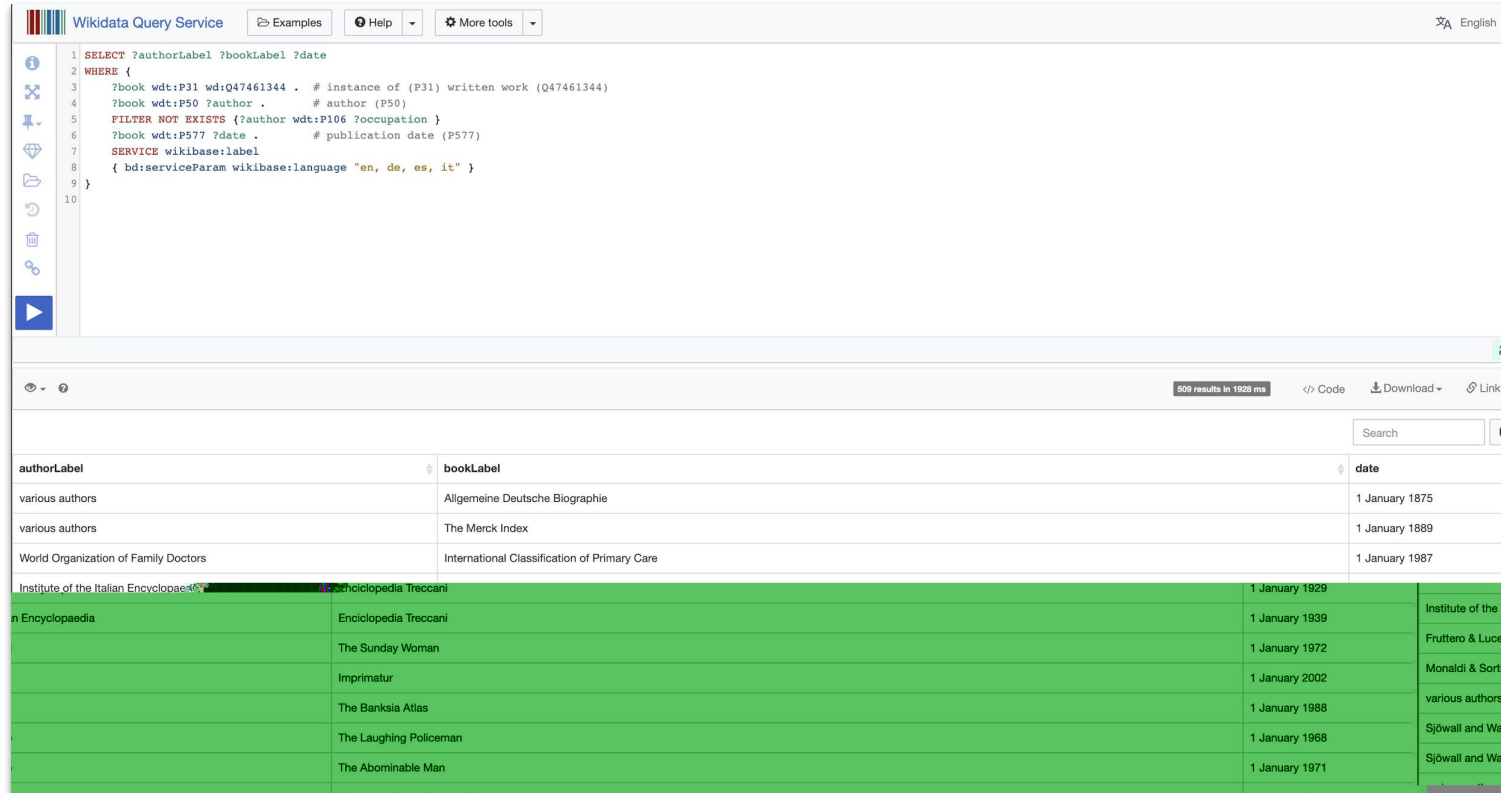
○

○

[query SPARQL endpoint](#)

3. Querying RDF(S) with SPARQL / 3.2 Complex Queries with SPARQL

- Example: which books are written by authors who don't have an occupation ?



The screenshot shows the Wikidata Query Service interface. The query is as follows:

```

1 SELECT ?authorLabel ?bookLabel ?date
2 WHERE {
3   ?book wdt:P31 wd:Q47461344 . # instance of (P31) written work (Q47461344)
4   ?book wdt:P50 ?author .      # author (P50)
5   FILTER NOT EXISTS {?author wdt:P106 ?occupation }
6   ?book wdt:P577 ?date .      # publication date (P577)
7   SERVICE wikibase:label
8   { bd:serviceParam wikibase:language "en, de, es, it" }
9 }
10

```

The results table shows the following data:

authorLabel	bookLabel	date
various authors	Allgemeine Deutsche Biographie	1 January 1875
various authors	The Merck Index	1 January 1889
World Organization of Family Doctors	International Classification of Primary Care	1 January 1987
Institute of the Italian Encyclopaedia	Enciclopedia Treccani	1 January 1929
Institute of the Italian Encyclopaedia	Enciclopedia Treccani	1 January 1939
	The Sunday Woman	1 January 1972
	Imprimatur	1 January 2002
	The Banksia Atlas	1 January 1988
	The Laughing Policeman	1 January 1968
	The Abominable Man	1 January 1971

[query SPARQL endpoint](#)

A detailed scientific illustration of various cephalopods. In the center, a squid is shown vertically with its head at the top, featuring large eyes and a long, tapering mantle. Surrounding the squid are several octopus tentacles, some curled and others extended, showing the characteristic suckers. The illustration is rendered in a classic scientific style with fine lines and some color washes.

More Complex SPARQL Queries

Next Lecture... "lrj lEjkg l..."

Picture References:

- [1] Benjamin Nowack, *The Semantic Web - Not a Piece of cake...*, at [bnode.org](http://bnode.org/blog/2009/07/08/the-semantic-web-not-a-piece-of-cake), 2009-07-08 , [CC BY 3.0]
<http://bnode.org/blog/2009/07/08/the-semantic-web-not-a-piece-of-cake>
- [2] British Crown vector illustration, [publicdomainvectors.org](https://publicdomainvectors.org/en/free-clipart/British-Crown-vector-illustration/12150.html), [Public Domain]
<https://publicdomainvectors.org/en/free-clipart/British-Crown-vector-illustration/12150.html>
- [3] Wikidata Logo, Planemad, [Public Domain]
<https://commons.wikimedia.org/wiki/File:Wikidata-logo-en.svg>
- [4] Ernst Haeckel, *Kunstformen der Natur* (1904), plate 54: Gamochonia, [Public Domain]
https://commons.wikimedia.org/wiki/File:Haeckel_Gamochonia.jpg